Uniform collection: unexecuted Lean source draft

Existing PR2381;26 October2026

No new theorem is verified. This prepares source while checkpoint upload authorization and token access remain pending. The file ends in `.lean.txt`, outside all proof-gate projects. No Lean elaboration, kernel check, axiom audit or native scientific execution occurs. Existing 208public/269audited declarations and17identities stay unchanged.

The intended bridge

The objective remains better useful relation yield and lower relevant solver degree for an actual twist, with a checked source-DLP reduction. The frozen uniform protocol still plans96arms,896targets and210policy pairs on the existing F43/F13 fixtures; no planned comparison has executed.

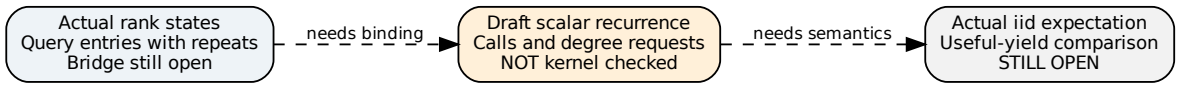
For coefficient space L, intended height is $b - \dim L$ and $N = r - 1$. List every improving successor once per query, *including repeated successors*. A stayer preserves the entire subspace, not only dimension. The handwritten draft exposes mass conservation, positive progress, strictly decreasing height and terminal boundary hypotheses.

Proposed algebraic statements, not checked results

Given an explicit first-step identity with total outcome cost c, let $A_F(L)$ be the sum of $F(L')$ over improving query outcomes, with repeats:

$$NF(L) = c + s(L)F(L) + A_F(L), \qquad F(L) = \frac{c + A_F(L)}{N - s(L)}.$$

The draft targets uniqueness by decreasing height and linear scaling. Calls use $c = N$. If D is the sum of the same named query degrees at every state, the proposed degree-request identity is $C(L) = (D/N)E(L)$. For one remaining rank, all improving successors terminate, giving the proposed special case $E = 1/p$ with $p = (N - s)/N$.



What does not follow. Larger empty-state expected rank gain alone does not establish smaller total collection calls. The comparison draft explicitly assumes no larger call count as well as smaller mean degree. Degree requests are not calibrated matrix work, total DLP cost or timing.

Remaining checks and launch hold

The first-step equation is a hypothesis, not a theorem about a random stopping time. Actual subspace/query/source bindings, iid sampling with replacement, absorption, integrability and probability semantics remain open. After checkpoint-only authorization, secure token setup and successful unmodified gate, activate a new pinned Lean project and retain all attempts through the unmodified gate/axiom audit. Do not overwrite this draft or older proofs. Native replay/controls and the exact experiment runner remain pending.

Source directory: `drafts/uniform-collection-20261026/`.
Draft file: `UniformCollection.lean.txt`.
Contract: `UNIFORM_COLLECTION_20261026_DRAFT_CONTRACT.md`. Dependencies target Lean4.34.1/Mathlib`d13f23b7...`.
The report uses a diagram because no new comparable scientific values exist. No new upload, credential reuse, paid resource, registry/benchmark row or canonical graph.